

Ayman Mahfuz

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EDUCATION

The University of Texas at Austin

Austin, TX

Bachelor of Science in Computer Science

May 2027

- **Concentration:** AI and ML — **GPA:** 3.5/4.0
- **Courses:** Data Structures, Algorithms, High Performance Computing, Natural Language Processing, Generative Visual Computing, Energy-Efficient Computing

EXPERIENCE

Arm

May 2025 – Aug 2026

Platform Validation Intern / Co-op – Software Engineering

- Reproduced silicon failures across **three CPU-platform validation** phases with a production **Python Bayesian optimizer** for **stress-ng** configurations; owned development and deployment.
- Guided stress-test selection with **CPU performance counters**, optimizing **cache-miss rate**, **memory bandwidth**, and **stall rate** to identify high-stress configurations.
- Earned internal patent-filing approval and a **12-month** co-op extension; presented production results to SVP and director leadership.

UT Austin – Texas Robotics & AI Lab

Jan 2025 – Jul 2026

Undergraduate Research Assistant – Robotics Software & Reinforcement Learning

- Accelerated policy-training iterations **2–3x** by profiling **GPU** utilization and removing training-pipeline bottlenecks; scaled experiments to **5M+ episodes**.
- Deployed **15 hierarchical reinforcement-learning** policies to **NAO robots** and built **C++/Python** training environments within a **400K+ line** codebase, contributing to the team's third-place finish at RoboCup 2025.

UT Austin – Center for Media Engagement

Aug 2023 – May 2025

Undergraduate Research Assistant – Software Engineering

- Built concurrent **Selenium/REST** ingestion workers with retry handling for a **250M+ row BigQuery dataset** containing **50M+ articles** and **70M+ comments**.
- Prevented duplicate experiment events and preserved round-robin assignments for **1,000+ participants** through deterministic assignment logic, event deduplication, and structured logging.

UT Austin – Oden Institute

Aug 2023 – Jan 2025

Undergraduate Research Assistant – ML Systems

- Improved MRI-segmentation Dice score by **12 percentage points** through **convolutional and transformer** model experiments with **five-fold validation** on **1,000+ scans** using **NVIDIA H100 GPUs**.
- Stabilized long-running **PyTorch** training and recovered failed jobs from checkpoints by resolving **GPU-memory**, **I/O**, and **mixed-precision** bottlenecks in an **Apptainer/SLURM** pipeline.

PROJECTS

HelmPM – Multi-Tenant Project Management Platform | *React, FastAPI, PostgreSQL*

- Built a multi-tenant platform managing **thousands of tickets across four organizations**, with a **React** frontend, **FastAPI** backend, **PostgreSQL** database, and automated **Linear/Jira** synchronization.
- Eliminated re-embedding of unchanged documents with **Merkle-tree indexing**, reducing API costs by **40%**; implemented **semantic**, **keyword**, and **exact-ID search** across documents and code.

253M-Parameter Language Model & Training Pipeline | *PyTorch, HPC*

- Implemented and pretrained a **253M-parameter transformer** on **600M tokens** in **PyTorch** using **RoPE**, **RMSNorm**, and **SwiGLU**.
- Supported multi-session **HPC** experiments with **mixed precision**, **gradient accumulation**, **checkpoint recovery**, and **evaluation**; implemented **supervised fine-tuning** and **direct preference optimization**.

LLM & Diffusion Inference Energy Benchmark | *Python, PyTorch, vLLM, SLURM*

- Measured **24–32% lower LLM inference energy use** with **FP8/INT8** and no measurable diffusion savings across **577 tests** spanning **nine models** and **55 configurations** on **GH200 GPUs**.
- Built a **Python/vLLM/SLURM** harness instrumented with **NVML** and **Zeus**; evaluated differences with **Welch's t-tests** and **Bonferroni correction** and interpreted results through **roofline analysis**.

SKILLS

Languages: Python, C++, Java, TypeScript

Backend/Web/Data: React, FastAPI, Flask, PostgreSQL, BigQuery, REST APIs, Pandas, NumPy, AWS

ML/Systems: PyTorch, Transformers, vLLM, LoRA/PEFT, PPO/MAPPO, RLHF/DPO, SLURM, Docker, Linux, profiling, MCP, Harness Engineering, Zeus/NVML, HPC, Distributed Systems